

NCR MANAGER · QUALITY MANAGEMENT SYSTEM

Workflow & State- Transition Definitions

Kinetic Task-Engine trees: risk-based routing, approvals, escalation timers and immutable audit history.

DOCUMENT

WORKFLOWS

VERSION

1.0

ISSUED

June 27, 2026

PLATFORM

Kinetic

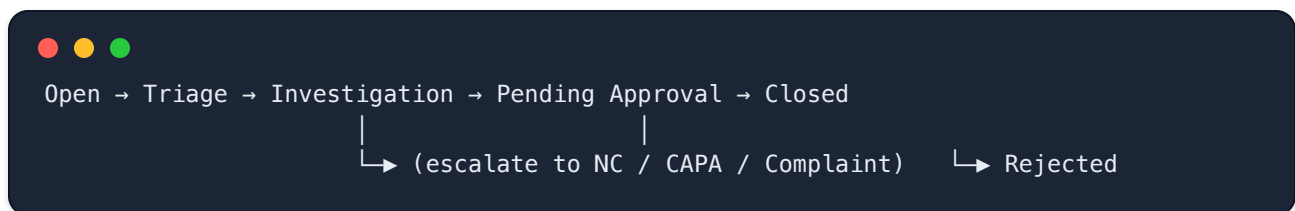
CONFIDENTIAL · CONTROLLED DOCUMENT

Workflows are implemented as Kinetic **Task Engine** trees triggered on submission create/update. The UI and server hold **no process logic** — they read state. Each tree applies risk-based routing, business-calendar due dates, reminder/escalation timers, parallel/sequential approvals, delegation with provenance, separation-of-duties guards, and writes every transition to `audit-trail`.

Common capabilities across all trees:

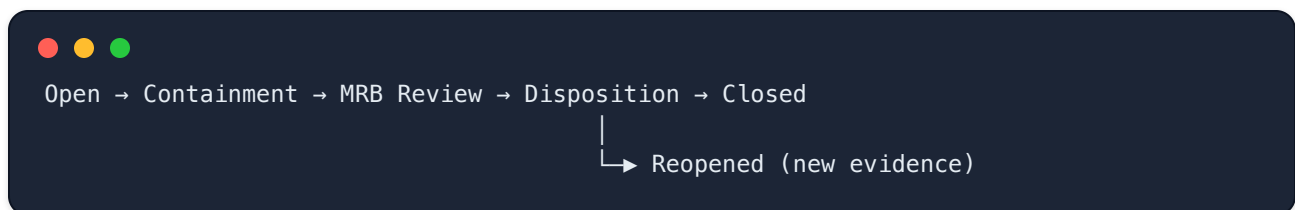
- **Risk-based routing** — High/Critical records require an extra approval level and an *independent* effectiveness/verification reviewer.
- **Due dates** — computed from severity/risk against a business calendar; stored on the record.
- **Reminders & escalation** — timer nodes notify owner before due, escalate to the manager team after.
- **Delegation** — reassignment captures delegator, delegate, reason and time (provenance in audit trail).
- **Reject / rework / cancel / reopen / supersede** — explicit transitions, each reason-bearing.
- **Immutable history** — transitions are append-only; records are amended, never silently overwritten.
- **Queues** — task assignment to individual / team / site / role; surfaced in *My Work*.

Quality Event



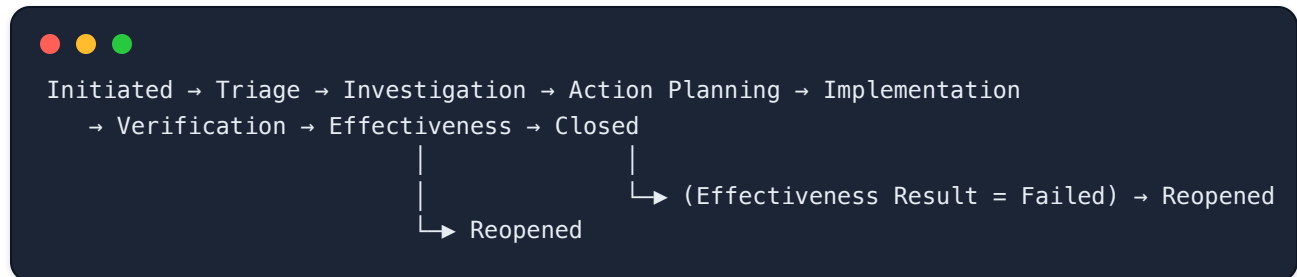
Triage sets severity/risk and disposition: close (no action), correct, or **escalate** (spawns NC, CAPA or Complaint and links bi-directionally). Confidential/anonymous flags route to a restricted queue.

Nonconformance / Material Review



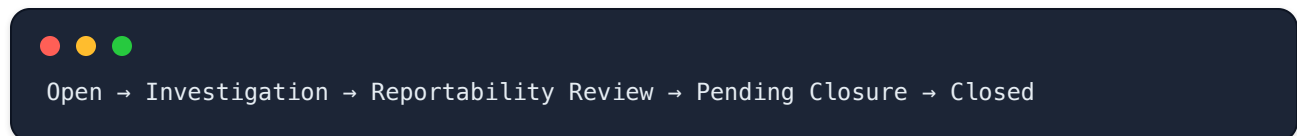
Disposition $\in$ {Use-As-Is, Rework, Scrap, Return to Supplier, Concession}; Use-As-Is and Concession require **MRB approval** (Quality Manager). Containment/segregation and quantity affected are snapshotted.

CAPA (the spine)



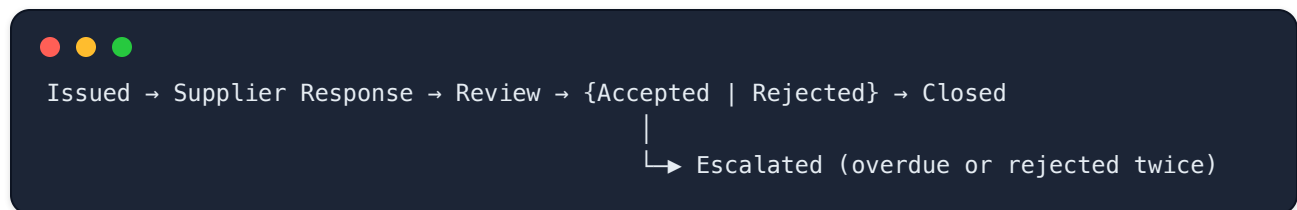
Guards before **Closed**: all `capa-actions` Completed **and** verified, effectiveness criteria defined and result = Passed, required signatures present (Effectiveness Approved by an *independent* reviewer + CAPA Closure Approved). Investigation method ∈ {Five Whys, Ishikawa, Fault Tree, Is/Is-Not, Human Factors, combinations}; multiple suspected/confirmed root causes supported.

Complaint & Vigilance



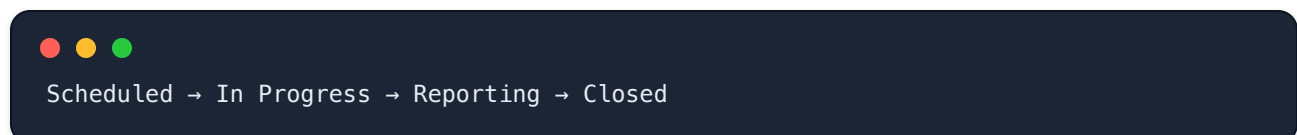
Reportability Review yields Reportable ∈ {Yes, No, Undetermined}; if Yes, a regulatory clock (MDR/MIR) and `Regulatory Due Date` are set and surfaced as overdue when breached. Duplicate detection links to the primary complaint; returned-product analysis and CAPA/risk links captured.

CAR / SCAR (supplier)



Supplier contacts respond via the isolated portal. Acceptance verifies effectiveness and applies a **supplier score impact**; rejection re-issues with a new due date; overdue auto-escalates.

Audit



Findings (`Major` | `Minor` | `Observation` | `OFI`) get response → remediation → verification; Majors may spawn a linked CAPA. Auditor independence enforced at assignment.

| Document & Change Control



Document: Draft → In Review → Approved → Effective → Obsolete

Change: Draft → Impact Assessment → Approval → Implementation → Verification → Closed

Approving an effective document/change with **Training Required = Yes** **auto-assigns training** to the affected role/site population. Effective-date and obsolete-version control maintained.

| Training & Competency



Assigned → In Progress → Completed (→ Overdue if past due; → re-assigned on Expiration)

Auto-assigned from document revisions, role changes, CAPAs and quality events. Overdue escalates; expiration triggers retraining. Effectiveness via test score / practical qualification.

| Risk (ISO 14971)



Open → Controlled → Monitoring

Initial risk = Severity × Probability; controls + verification reduce to residual; benefit-risk decision recorded. Post-production signals (complaints, CAPAs, NCs) re-open assessment.

| Management Review



Draft → In Progress → Approved

Aggregates audit results, complaints, CAPA performance, supplier performance, process metrics, regulatory changes, risks, prior actions and improvement opportunities into a governed package, approved by signature.

| Reusable handlers (integration touch-points, design)

ERP (lot/inventory hold) · MES (process exception → event) · PLM (design/doc change) · HRIS (role → training) · LIMS (test result → NC) · Calibration (OOT → equipment + affected-product) · Supplier

portal · Email/notification · Document storage · Analytics/data platform. Each is a versioned task handler invoked from the trees above; failures are retried and logged with correlation IDs.